

Answers to Math 124 Autumn 2023 Final Exam

$$1. \quad (a) \quad \lim_{x \rightarrow 2} \frac{\sqrt{x^2 + 5} - 3}{x^2 - 2x} \cdot \frac{\sqrt{x^2 + 5} + 3}{\sqrt{x^2 + 5} + 3} = \lim_{x \rightarrow 2} \frac{x^2 - 4}{(x^2 - 2x)(\sqrt{x^2 + 5} + 3)} = \lim_{x \rightarrow 2} \frac{x + 2}{x(\sqrt{x^2 + 5} + 3)} = \frac{1}{3}$$

$$(b) \quad \lim_{x \rightarrow \pi/2} \frac{1 - \sin x}{x - \pi/2} \stackrel{\text{LH}}{=} \lim_{x \rightarrow \pi/2} \frac{-\cos x}{1} = 0$$

$$(c) \quad \text{From } y = (\sin x)^{\tan x},$$

$$\ln y = (\tan x)(\ln(\sin x)) = \frac{\ln(\sin x)}{\cot(x)}.$$

Since

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{\ln(\sin x)}{\cot(x)} \stackrel{\text{LH}}{=} \lim_{x \rightarrow 0^+} \frac{\frac{\cos x}{\sin x}}{-\csc^2(x)} = - \lim_{x \rightarrow 0^+} \sin(x) \cdot \cos(x) = 0$$

$$\lim_{x \rightarrow 0^+} y = e^0 = 1$$

$$2. \quad (a) \quad f(x) = \frac{x \sin(x^3)}{1 + e^{4x}}$$

$$f'(x) = \frac{(\sin(x^3) + x \cdot 3x^2 \cdot \cos(x^3))(1 + e^{4x}) - (x \sin(x^3))(4e^{4x})}{(1 + e^{4x})^2}$$

$$(b) \quad g(x) = \tan(a + \ln(b + e^{cx})) \text{ where } a, b, c \text{ are constants.}$$

$$g'(x) = \sec^2(a + \ln(b + e^{cx})) \cdot \frac{ce^{cx}}{b + e^{cx}}$$

$$(c) \quad k(x) = \arctan\left(\frac{2x - 1}{4 + 7x}\right)$$

$$k'(x) = \frac{1}{1 + \left(\frac{2x-1}{4+7x}\right)^2} \cdot \left(\frac{2(4 + 7x) - 4(2x - 1)}{(4 + 7x)^2}\right)$$

$$3. \quad (a) \quad f'(0) = -1$$

$$(b) \quad w'(7) = -8$$

$$(c) \quad u'(0) = -\frac{4}{5}$$

$$(d) \quad x = 3, -4.65$$

$$(e) \quad (1, 3) \text{ and } (4, \infty)$$

4. Implicit differentiation:

$$4(x^2 + y^2)(2x + 2yy') = 25(2x - 2yy').$$

so

$$y' = \frac{50x - 8x(x^2 + y^2)}{50y + 8y(x^2 + y^2)}$$

tangent is vertical when the denominator equals zero:

$$0 = 50y + 8y(x^2 + y^2) = y(50 + 8(x^2 + y^2))$$

since $50 + 8(x^2 + y^2) > 0$ we must have $y = 0$ so using the curve equation

$$2(x^2)^2 = 25(x^2).$$

so $x^2(2x^2 - 25) = 0$. Since $x \neq 0$, we get $x = \pm\sqrt{5/2}$ so two points $(\pm\sqrt{5/2}, 0)$.

5. (a) From

$$\frac{dy}{dx} = \frac{3t^2 + 8t}{e^t + 1}$$

the tangent is horizontal when $3t^2 + 8t = 0$ so $t = 0$ or $t = -8/3$. The tangent is never vertical since $e^t + 1 \neq 0$.

- (b) From

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \frac{3t^2+8t}{e^t+1}}{\frac{dx}{dt}} = \frac{e^t(-3t^2 - 2t + 8) + 6t + 8}{(e^t + 1)^3}$$

at $t = 0$

$$\left. \frac{d^2y}{dx^2} \right|_{t=0} = \frac{1(8) + 8}{(2)^3} > 0$$

so concave up and at $t = -8/3$

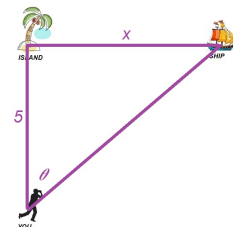
$$\left. \frac{d^2y}{dx^2} \right|_{t=-3/8} = \frac{e^{-3/8}(-56/8) - 8}{(e^{-3/8} + 1)^3} < 0$$

so concave down.

6. Differentiate $\tan \theta = x/5$ and plug in values:

$$\sec^2 \theta \frac{d\theta}{dt} = \frac{1}{5} \frac{dx}{dt} \quad \text{so} \quad \sec^2(\pi/6) \frac{5}{4} = \frac{1}{5} \frac{dx}{dt}$$

to get $dx/dt = 25/3$ miles per hour.



7. Using similar triangles $b = \frac{2a}{a-1}$ so the area of the triangle is $A = \frac{a^2}{a-1}$ with domain $a > 1$.
From

$$A' = \frac{a^2 - 2a}{(a-1)^2}$$

we get the critical number $a = 2$ (since $a = 0$ is not possible). Then $b = 4$. To check it is minimum:

$$A'' = \frac{(2a-2)(a-1)^2 - (a^2-2a) \cdot 2(a-1)}{(a-1)^4} = \frac{-2}{(a-1)^2} < 0$$

8. (a) $\lim_{x \rightarrow 0^-} \frac{e^x}{x} = -\infty$ and $\lim_{x \rightarrow 0^+} \frac{e^x}{x} = \infty$

- (b) $\lim_{x \rightarrow \infty} \frac{e^x}{x} \stackrel{\text{LH}}{=} \lim_{x \rightarrow \infty} \frac{e^x}{1} = \infty$ and $\lim_{x \rightarrow -\infty} \frac{e^x}{x} = 0$ so $y = 0$ is a horizontal asymptote on the left as $x \rightarrow -\infty$.

- (c) From $f'(x) = \frac{e^x(x-1)}{x^2}$ the function is decreasing on $(-\infty, 0)$ and $(0, 1)$.

- (d) From

$$f'(x) = \frac{e^x(x^2 - 2x + 2)}{x^3}$$

the graph is concave up for $x > 0$ and concave down for $x < 0$.

